Allen Tu

Ph.D. Student, Computer Science — University of Maryland, College Park
3D/4D Digital Twins, World Models, Generative AI

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Selected Publications

- **Allen Tu***, H. Ying*, A. Hanson, Y. Lee, T. Goldstein, and M. Zwicker, 'SpeeDe3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping'. *CVPR*, 2026.
- P. Asthana, A. Hanson, **Allen Tu**, T. Goldstein, M. Zwicker, and A. Varshney, 'SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting'. *CVPR*, 2026.
- **Allen Tu**, K. Narayan, J. Gleason, J. Xu, M. Meyn, and V. Patel, 'TransFIRA: Transfer Learning for Face Image Recognizability Assessment'. *FG*, 2026 (**Oral Presentation; Invited to submit an extended version to the IEEE T-BIOM Special Issue on the Best-Ranked Articles of IEEE FG 2026**).
- A. Hanson*, **Allen Tu***, V. Singla, M. Jayawardhana, M. Zwicker, and T. Goldstein, 'PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting'. *CVPR*, 2025.
- A. Hanson, **Allen Tu**, G. Lin, V. Singla, M. Zwicker, and T. Goldstein, 'Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives'. *CVPR*, 2025.

* denotes equal contribution.

Education

University of Maryland, College Park — College Park, MD
Ph.D. Student, Computer Science — *January 2025 – May 2028 (Expected)*

- **Advisors:** Professors Tom Goldstein, Matthias Zwicker

M.S. in Computer Science — *August 2023 – December 2024*

B.S in Computer Science, Minor in Statistics — *August 2019 – December 2022*

Research Experience

Amazon — Bellevue, WA
Applied Science Intern — *June 2026 – Present*

- Researching 3D digital twins, world models, and agentic pipelines for physical AI simulation in the Last Mile Delivery and Automation org
- Created a method that reconstructs simulation-ready 3D digital twins directly from fisheye imagery, combining photorealistic Gaussian Splatting over the camera's full field of view with an aligned mesh for collisions [1]
- Constructing scalable pipelines that produce photorealistic, metric-scale 3D digital twin simulation environments from large-scale passive data collection

University of Maryland Institute of Advanced Computer Studies — College Park, MD
Graduate Research Assistant — *August 2023 – Present*

- Currently researching latent world models, unconstrained 3D/4D reconstruction, and generative modeling under Professors Tom Goldstein and Matthias Zwicker
- Published papers on efficient 3D/4D Gaussian Splatting [4, 7, 8], 3D reconstruction from degraded and fisheye imagery [1, 2, 5], conditioned CAD model generation [3], and biometric recognition [6]
- Developed scalable 3D reconstruction pipelines deployed for unconstrained novel-view synthesis in real-world environments under the IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA) program

Systems & Technology Research Arlington, VA
Computer Vision Research Intern *May 2025 – January 2026*

- Engineered multimodal biometric systems fusing face, body, and gait recognition for robust identification under severe conditions in the IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR) program
- Proposed an explainable transfer-learning method for face image quality assessment (FIQA), enabling recognizability-aware probe filtering and weighted template aggregation for face and body recognition [6]
- Demonstrated state-of-the-art template-based recognition on the proprietary BRIAR surveillance benchmark, improving TAR from 0.43 to 0.87 at 1e-3 FMR using an encoder trained solely on public data [6]

Computer Vision Research Intern *May 2024 – August 2024*

- Achieved 0.86 correlation with ground-truth probe–to–gallery similarity and state-of-the-art ERC performance for face image quality estimation (FIQA), outperforming prior methods on image-level evaluation [6]
- Designed a cluster-and-aggregate network that filters low-importance frame embeddings and fuses the remaining informative ones into a single video template, improving face recognition accuracy by up to 12%

Computer Vision Research Co-op *January 2023 – August 2023*

- Implemented a mixed-voting ensemble that improved open-set search performance by 24% for face recognition, 39% for body recognition, and 49% for fused multimodal biometric recognition
- Trained a self-supervised Barlow Twins model robust to challenges like extreme distance variation and atmospheric interference, improving face recognition accuracy by 52% in severe operating conditions

Computer Vision Research Intern *June 2022 – August 2022*

- Integrated pose estimation, neural novel-view synthesis, semantic segmentation, and generative inpainting to synthesize realistic cross-subject garment transfers with strong biometric identity preservation
- Augmented real training data with garment-transfer images to overcome limited garment diversity and train clothing-invariant whole-body recognition models, improving robustness to cross-garment variation

University of Maryland Department of Computer Science College Park, MD
Undergraduate Researcher *January 2021 – December 2022*

- Analyzed 2.3 million dataset–visualization pairs to identify bias in machine learning visualization recommenders, linking model behavior to user-driven design patterns and proposing data-centric mitigation strategies [9]

Research Projects

Last Hundred Yards
Amazon Last Mile Delivery and Automation *June 2026 – Present*
Unconstrained 3D reconstruction and novel view synthesis in challenging environments.

- Simulation-ready 3D digital twins [1]
- 3D generation and reconstruction
- World models and video generation
- Agentic pipelines

IARPA Walk-through Rendering from Images of Varying Altitude (WRIVA)
University of Maryland Institute of Advanced Computer Studies *August 2023 – November 2026*
Unconstrained 3D reconstruction and novel view synthesis in challenging environments.

- Efficient rendering, compression, and training methods for 3D and 4D Gaussian Splatting (3DGS/4DGS) [4, 7, 8]
- Multi-view consistent super-resolution for training 3DGS with low-resolution imagery [5]
- Image, video, and multi-view diffusion model priors for sparse-view 3D reconstruction
- Uncertainty quantification algorithms for 3DGS and Neural Radiance Fields (NeRF) [7]

IARPA Biometric Recognition and Identification at Altitude and Range (BRIAR)
Systems & Technology Research *June 2022 – January 2026*

Multimodal, opportunistic fusion of face, body, and gait information in severe operational conditions.

- Transfer Learning for Face Image Recognizability Assessment [6]
- Learned Frame Feature Aggregation for Face Recognition with Low-Quality Video
- Operating Condition-Invariant Barlow Twins and Multimodal Ensembling
- Style-Based Clothing Transfer for Robust Body Representation Learning

Honors & Awards

FG 2026 Doctoral Consortium: *Selective Computation for Reliable and Scalable Vision Systems*

Amazon Many Minds, Smarter Outcomes Award: Recognized for work on bird's-eye-view perception for quadruped robot teleoperation

University of Maryland Travel Awards: Jacob K. Goldhaber Travel Grant, CMSC PhD Travel Grant, ICSSA

Service

SPAR-3D: Security, Privacy, and Adversarial Robustness in 3D Generative Vision Models
Workshop Co-organizer *CVPR 2026*

- Co-organized the first CVPR workshop on security, privacy, and robustness in 3D generative vision models, coordinating peer review, securing $1K in industry sponsorship, and moderating keynote and Q&A sessions

Reviewer: IEEE TVCG, SIGGRAPH Asia, IEEE TBIOM, CVPR Workshops, FG, IEEE TCSVT

Conference Service: Presented 5 accepted papers (1 oral, 4 posters) on behalf of authors unable to attend FG 2026

Organizations: IEEE, ACM, Computer Vision Foundation, IEEE Biometrics Council

Teaching Experience

Peer Research Mentor College Park, MD
FIRE: Capital One Machine Learning *January 2021 – December 2022*

- Mentored 80 undergraduate students across two cohorts in a selective multi-semester machine learning research program, guiding students from foundational training to independent research project execution
- Designed and delivered a project-driven curriculum spanning data processing, model development, training, evaluation, and deployment using modern deep learning frameworks and recent research advances
- Advised student teams end-to-end on collaborative research projects, leading to presentations in areas including text-to-audio generation, image super-resolution, and face attribute recognition

Technical Skills

Languages: Python, CUDA, C/C++, C#, Java, SQL, Shell

Frameworks & Libraries: PyTorch, TensorFlow, NumPy, SciPy, OpenCV, scikit-learn, COLMAP, Open3D, PhysX

Specializations: 3D/4D Reconstruction, 3D Gaussian Splatting, Neural Radiance Fields (NeRF), Differentiable Rendering, Diffusion Models, Multimodal Biometrics, Uncertainty Quantification, Vision Transformers, GPU Programming

PUBLICATIONS

For the full and up-to-date publication list, see Google Scholar.

1. **Allen Tu**, Meher Gitika Karumuri, Shifan Zhu, Hailing Wang, Yen-Liang Lin, David Held, Matthias Zwicker, Tom Goldstein, 'GOFish: Full Field-of-View Gaussian Splatting and Mesh Extraction'. *Under review*, 2026.
2. Kevin David Hayes, Alex Hanson, **Allen Tu**, Pranav Asthana, Matthias Zwicker, Tom Goldstein, 'RIDE-Splat: Reflection Removal for In-the-Wild Driving Environments'. *Under review*, 2026.
3. Haiyang Ying, **Allen Tu**, Jiaye Wu, Tom Goldstein, and Matthias Zwicker, 'UniBRep: Learning Unified Geometry and Topology for Image-conditioned B-Rep Generation'. *Under review*, 2026.
4. **Allen Tu***, Haiyang Ying*, Alex Hanson, Yonghan Lee, Tom Goldstein, and Matthias Zwicker, 'SpeeDe3DGS: Speedy Deformable 3D Gaussian Splatting with Temporal Pruning and Motion Grouping'. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2026, pp. 26083–26093.
5. Pranav Asthana, Alex Hanson, **Allen Tu**, Tom Goldstein, Matthias Zwicker, and Amitabh Varshney, 'SplatSuRe: Selective Super-Resolution for Multi-view Consistent 3D Gaussian Splatting'. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2026, pp. 11840–11849.
6. **Allen Tu**, Kartik Narayan, Joshua Gleason, Jennifer Xu, Matthew Meyn, and Vishal M. Patel, 'TransFIRA: Transfer Learning for Face Image Recognizability Assessment'. *Proceedings of the IEEE 20th International Conference on Automatic Face and Gesture Recognition (FG)*, 2026, pp. 1–9, doi: 10.1109/FG67764.2026.11556959 (**Oral Presentation; Invited to submit an extended version to the IEEE Transactions on Biometrics, Behavior, and Identity Science (T-BIOM) Special Issue on the Best-Ranked Articles of IEEE FG 2026**).
7. Alex Hanson*, **Allen Tu***, Vasu Singla, Mayuka Jayawardhana, Matthias Zwicker, and Tom Goldstein, 'PUP 3D-GS: Principled Uncertainty Pruning for 3D Gaussian Splatting'. *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, 2025, pp. 5949–5958, doi: 10.1109/CVPR52734.2025.00558.
8. Alex Hanson, **Allen Tu**, Geng Lin, Vasu Singla, Matthias Zwicker, and Tom Goldstein, 'Speedy-Splat: Fast 3D Gaussian Splatting with Sparse Pixels and Sparse Primitives'. *Proceedings of the Computer Vision and Pattern Recognition Conference (CVPR)*, 2025, pp. 21537–21546, doi: 10.1109/CVPR52734.2025.02006.
9. **Allen Tu**, Priyanka Mehta, Alexander Wu, Nandhini Krishnan, and Amar Mujumdar, 'Addressing Bias in Visualization Recommenders by Identifying Trends in Training Data: Improving VizML Through a Statistical Analysis of the Plotly Community Feed'. *arXiv preprint arXiv:2203.04937*, 2022.

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